Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

Course Title: Basic Electrical and Electronic Circuit Lab

Course Code: 07131102W

Experiment 09

Design and Verification of Basic Logic Gates (AND, OR, NOT) Using BJTs

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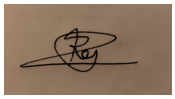
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Design and Verification of Basic Logic Gates (AND, OR, NOT) Using BJTs"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit Lab (07131102W)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

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Contents

Declaration **i**

List of Figures **iii**

List of Tables **iv**

Objective **1**

Theory **1**
- The BJT as a Switch . . . 1
- Resistor-Transistor Logic (RTL) . . . 1
- Logic Gate Configurations . . . 1

Apparatus and Components **1**

Experimental Procedure **2**

Data Collection **2**

Result Analysis **2**

Conclusion **3**

References **3**

Appendix: Signed Data Sheets **4**

List of Figures

1 Instructor-verified experimental data collection sheet (Page 1). 4
2 Instructor-verified experimental data collection sheet (Page 2). 5

List of Tables

1 List of Equipment and Components Used 1
2 AND Gate Truth Table and Measurements 2
3 OR Gate Truth Table and Measurements 2
4 NOT Gate Truth Table and Measurements 2

Objective

- To understand the working principle of Bipolar Junction Transistors (BJT) operating in switching modes (cutoff and saturation).
- To design and implement fundamental digital logic gates (AND, OR, NOT) using Resistor-Transistor Logic (RTL) configurations.
- To experimentally verify the truth tables of these logic gates by measuring input and output voltage levels.

Theory

The BJT as a Switch

In digital electronics, Bipolar Junction Transistors (BJTs) are utilized as solid-state switches by driving them between two extreme regions: cutoff ($I_C \approx 0$, acting as an open switch) and saturation ($V_{CE(sat)} \approx 0.2\,\text{V}$, acting as a closed switch).

Resistor-Transistor Logic (RTL)

Resistor-Transistor Logic (RTL) builds digital circuits using resistors as the input network and bipolar junction transistors as switching elements.

Logic Gate Configurations

- **AND Gate:** Formed by connecting two NPN transistors in series. Both must saturate simultaneously to drive the output high.
- **OR Gate:** Formed by connecting two NPN transistors in parallel. If either or both receive a high input, they conduct and drive the output high.
- **NOT Gate (Inverter):** Utilizes a single BJT in a common-emitter configuration. When the input is high, the transistor saturates, pulling the output low; when the input is low, the transistor cuts off, pulling the output high via the collector resistor.

Apparatus and Components

Table 1: List of Equipment and Components Used

Item Description	Specification / Rating	Quantity
DC Regulated Power Supply	5V or 6V	1
Digital Multimeter (DMM)	Standard	1
Solderless Breadboard	Standard Tie-Point	1
NPN Transistors	BC547 or 2N2222	As required
Resistors	$1\,\text{k}\Omega$, $4.7\,\text{k}\Omega$, $10\,\text{k}\Omega$, $220\,\Omega$, $330\,\Omega$	Various
Push Buttons / Switches	DIP switches or push buttons	As required
Connecting Wires	Jumper Wires	As required

Experimental Procedure

1. Construct the AND gate circuit using series NPN transistors, connect power, and apply inputs sequentially to record output voltages.
2. Construct the OR gate circuit using parallel NPN transistors, apply input combinations, and measure the output response.
3. Construct the NOT gate inverter circuit, apply high and low input states, and measure output voltage levels.

Data Collection

Table 2: AND Gate Truth Table and Measurements

Input A (V)	Input B (V)	Measured Output (V)	Logical Output State
0V (Logic 0)	0V (Logic 0)	1.15 V	0 (OFF)
0V (Logic 0)	5V (Logic 1)	1.80 V	0 (OFF)
5V (Logic 1)	0V (Logic 0)	1.20 V	0 (OFF)
5V (Logic 1)	5V (Logic 1)	5.34 V	1 (ON)

Table 3: OR Gate Truth Table and Measurements

Input A (V)	Input B (V)	Measured Output (V)	Logical Output State
0V (Logic 0)	0V (Logic 0)	0.001 V	0 (OFF)
0V (Logic 0)	5V (Logic 1)	4.43 V	1 (ON)
5V (Logic 1)	0V (Logic 0)	4.38 V	1 (ON)
5V (Logic 1)	5V (Logic 1)	4.42 V	1 (ON)

Table 4: NOT Gate Truth Table and Measurements

Input A (V)	Measured Output (V)	Logical Output State
0V (Logic 0)	5.12 V	1 (ON)
5V (Logic 1)	0.016 V	0 (OFF)

Result Analysis

The experimental measurements successfully validated the logical truth tables for the fundamental RTL AND, OR, and NOT gate configurations. For the AND gate, a high logical output was observed exclusively when both inputs were held at logic high (5 V). For the OR gate, a high output state was produced whenever at least one input received a high signal. The NOT gate configuration correctly inverted the input state, yielding a high output of 5.12 V for a low input and pulling the output down to 0.016 V when a high input saturated the BJT switch. Minor variations from ideal rail voltages reflect transistor saturation drops and component tolerances.

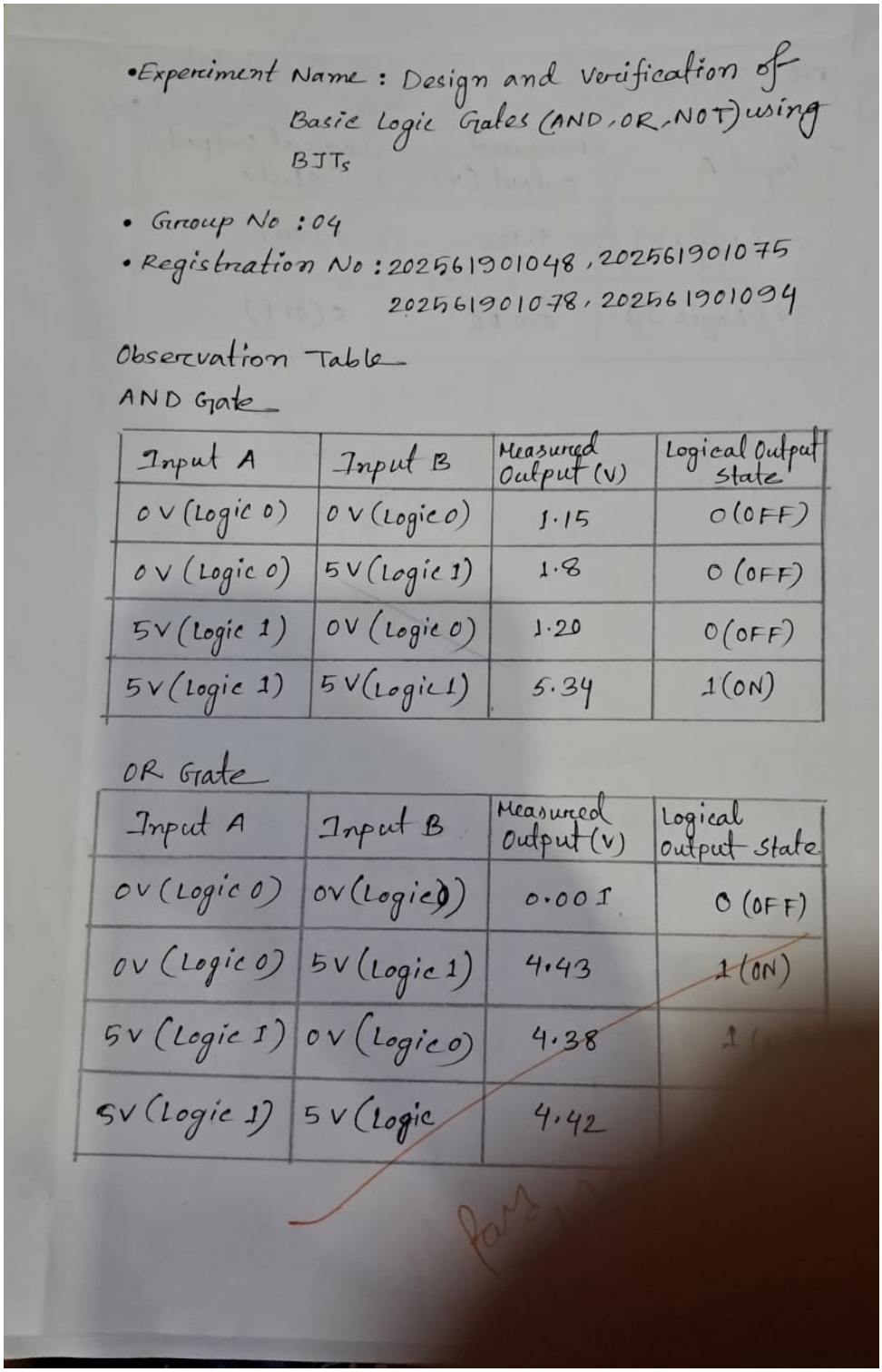
Conclusion

The fundamental Resistor-Transistor Logic (RTL) gates were successfully designed, constructed, and tested using bipolar junction transistors. Empirical verification of the truth tables confirmed that BJTs operate effectively as digital switches. Furthermore, quantitative output voltage measurements provided clear insight into practical digital state levels and switching dynamics.

References

1. Boylestad, R. L., & Nashelsky, L. (2013). *Electronic Devices and Circuit Theory* (11th ed.). Pearson.
2. Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits* (6th ed.). McGraw-Hill Education.
3. Sedra, A. S., & Smith, K. C. (2014). *Microelectronic Circuits* (7th ed.). Oxford University Press.

Appendix: Signed Data Sheets



•Experiment Name : Design and Verification of Basic Logic Gates (AND, OR, NOT) using BJTs

• Group No : 04

• Registration No : 202561901048, 202561901075
202561901078, 202561901094

Observation Table

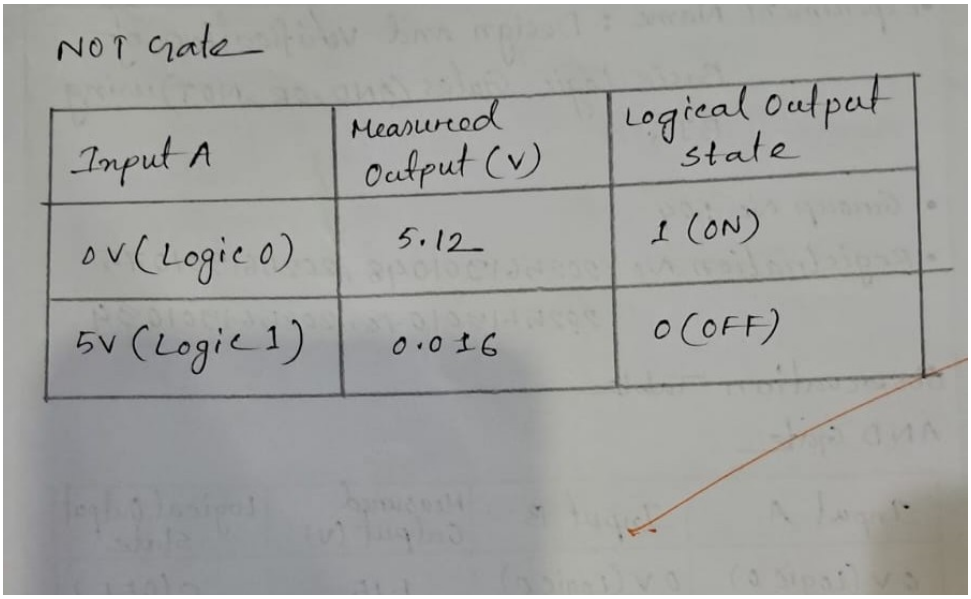
AND Gate

Input A	Input B	Measured Output (V)	Logical Output State
0 V (Logic 0)	0 V (Logic 0)	1.15	0 (OFF)
0 V (Logic 0)	5 V (Logic 1)	1.8	0 (OFF)
5 V (Logic 1)	0 V (Logic 0)	1.20	0 (OFF)
5 V (Logic 1)	5 V (Logic 1)	5.34	1 (ON)

OR Gate

Input A	Input B	Measured Output (V)	Logical output state
0 V (Logic 0)	0 V (Logic 0)	0.001	0 (OFF)
0 V (Logic 0)	5 V (Logic 1)	4.43	1 (ON)
5 V (Logic 1)	0 V (Logic 0)	4.38	1
5 V (Logic 1)	5 V (Logic	4.42	

Figure 1: Instructor-verified experimental data collection sheet (Page 1).


NOT Gate

Input A	Measured Output (V)	Logical Output State
0V (Logic 0)	5.12	1 (ON)
5V (Logic 1)	0.016	0 (OFF)

Figure 2: Instructor-verified experimental data collection sheet (Page 2).